University of Waterloo
Department of Mechanical and Mechatronics Engineering

ECE 140 – Linear Circuits

Final Examination – Winter 2025

SOLUTIONS

Date: April 9, 2025

Instructors: Section 1 – Derek Wright
Section 2 – Mike Cooper-Stachowsky

- **Exam duration: 150 minutes**
- **Closed book, double-sided letter-sized crib sheet allowed**
- **Clearly indicate final answers**
- **Show your work for part marks**
- **Make and clearly state any necessary assumptions**
- **Scientific calculators allowed (graphing is permitted)**

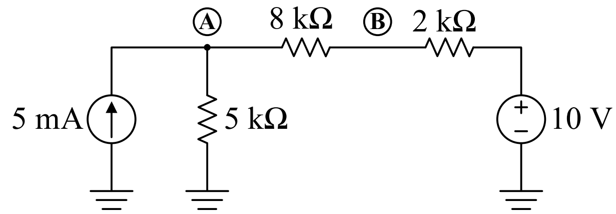
Question	Marks
Q1	39
Q2	10
Q3	8
Q4	10
Q5	10
Q6	6
Total	**83**

Question 1 – Short Questions

39 Marks

Part a) [4 Marks]

Determine the voltages at nodes A and B, V_A and V_B.



$V_A =$	20 V
$V_B =$	12 V

KCL @ A:

$$5 = \frac{V_A}{5} + \frac{V_A - V_B}{8}$$
$$200 = 13V_A - 5V_B$$

KCL @ B:

$$\frac{V_B - V_A}{8} + \frac{V_B - 10}{2} = 0$$
$$5V_B - V_A = 40$$

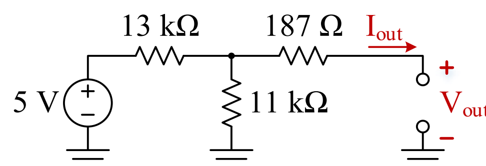
Solving:

$$200 = 13V_A - (40 + V_A)$$
$$240 = 12V_A$$
$$\therefore V_A = 20\text{ V}$$

$$5V_B - 20 = 40$$
$$\therefore V_B = 12\text{ V}$$

Part b) [3 Marks]

Determine V_{out} and I_{out}.



$V_{out} =$	2.29 V
$I_{out} =$	0 A

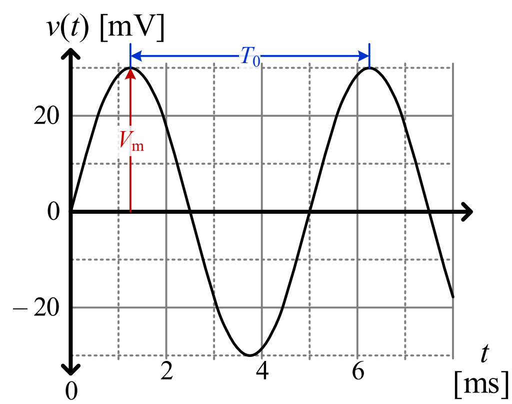
$I_{out} = 0$ A since output open-circuit.

$$V_{out} = 5\frac{11}{11 + 13} = 2.29\text{ V}$$

No voltage drop across 187 Ω resistor since $I_{out} = 0$ A.

Part c) [6 Marks]

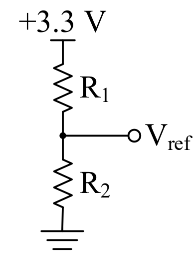
Given the oscilloscope plot of $v(t)$, state the signal's peak voltage (the amplitude, $V_{pk} = V_m$), RMS voltage (V_{RMS}), period (T_0), and frequency in Hz (f_0) and rad/s (ω_0). Using the peak voltage (not RMS), write the phasor representation for this sinusoidal waveform, $\boldsymbol{V}$, in polar notation, including the correct units.



$V_m =$	30 mV$_{pk}$
$V_{RMS} =$	$V_m/\sqrt{2} = 21.2$ mV$_{RMS}$
$T_0 =$	5 ms
$f_0 =$	$1/T_0 = 200$ Hz
$\omega_0 =$	$2\pi f_0 = 1.26$ krad/s
$\boldsymbol{V} =$	$30\angle -90°$ mV

Part d) [4 Marks]

Choose R_1 and R_2 so that $V_{ref} = 1.2$ V and the total power dissipated by R_1 and R_2 does not exceed 1 mW. Demonstrate through calculation that the power requirement is met.



$R_1 =$	$1.75R_2$ (6.93 kΩ min)
$R_2 =$	3.96 kΩ min

Voltage constraint:

$$V_{ref} = 3.3\frac{R_2}{R_1 + R_2}$$

$$1.2(R_1 + R_2) = 3.3R_2$$

$$R_1 = \frac{(3.3 - 1.2)R_2}{1.2} = \frac{2.1}{1.2}R_2 = 1.75R_2$$

Power constraint:

$$P_{Total} = P_{R_1} + P_{R_2} = \frac{V_{Total}^2}{R_{Total}} = \frac{3.3^2}{R_1 + R_2} \leq 1\text{ mW}$$

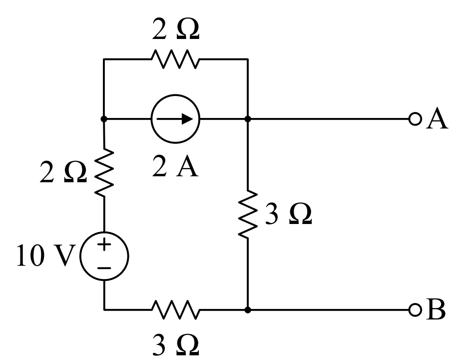
$$\therefore \frac{3.3^2}{1\text{m}} \leq R_1 + R_2$$

$$R_1 + R_2 \geq 10.89\text{ k}\Omega$$

At the lower limit, this leads to $R_1 = 6.93$ kΩ, and $R_2 = 3.96$ kΩ, so anything larger than these values that maintains the ratio is correct. It is also acceptable to choose a reasonable starting value and then demonstrate afterwards that the power requirement is met. For example, $R_2 = 10$ kΩ gives $R_1 = 17.5$ kΩ, then show that $P_{Total} \leq 1$ mW.

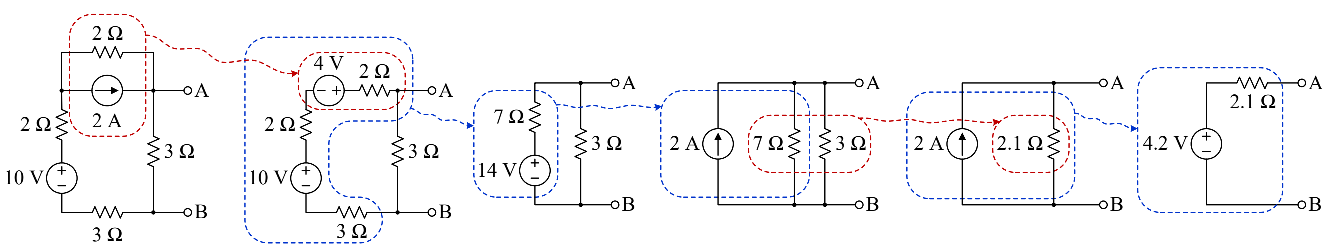
Part e) [3 Marks]

Determine the Thevenin equivalent circuit parameters, V_{Th} and R_{Th}, for node A with respect to node B.



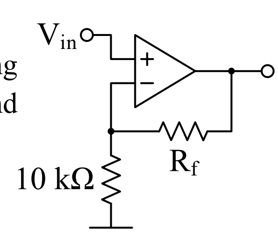
$V_{Th} =$	4.2 V
$R_{Th} =$	2.1 Ω

Use whatever approach you're most comfortable with. Here is simplification via source transforms:



Part f) [4 marks]

Determine the resistance R_f, to achieve a closed-loop gain $A_{ideal} = 20$ [V/V]. Assuming that **both resistors** have a ±5% tolerance, determine the worst-case maximum and minimum gains, A_{max} and A_{min}.



$R_f =$	190 kΩ
$A_{max} =$	22 V/V
$A_{min} =$	18.2 V/V

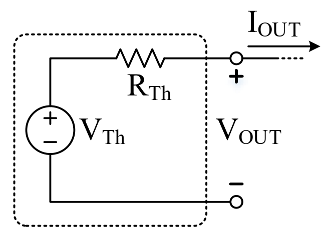
Noninverting amplifier:

$$A = 1 + \frac{R_f}{R_i}, \therefore R_f = (A - 1)R_i = (20 - 1) \times 10\text{k} = 190\text{ k}\Omega$$

$$A_{max} = 1 + \frac{R_{f,max}}{R_{i,min}} = 1 + \frac{1.05 \times 190}{0.95 \times 10} = 22\text{ V/V},$$

$$A_{min} = 1 + \frac{R_{f,min}}{R_{i,max}} = 1 + \frac{0.95 \times 190}{1.05 \times 10} = 18.2\text{ V/V}.$$

Part g) [4 marks]



A lab voltage supply is modelled as a Thevenin-equivalent circuit, as shown. When two different loads are connected one after the other (not shown), the following two operating points are observed:

- $V_{OUT} = 3.298\text{ V} \;@\; I_{OUT} = 30\text{ mA}$
- $V_{OUT} = 3.286\text{ V} \;@\; I_{OUT} = 270\text{ mA}$

Determine the Thevenin-equivalent voltage and resistance of the output.

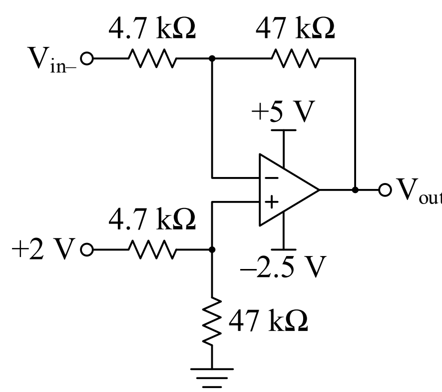
$V_{Th} =$	3.300 V
$R_{Th} =$	50 mΩ

$$R_{Th} = \frac{\Delta V_{OUT}}{\Delta I_{OUT}} = \frac{12\text{ mV}}{240\text{ mA}} = 50\text{ m}\Omega$$

If this wasn't apparent, you could use the KVL equation below with both data points to solve two equations in two unknowns. V_{Th} can be determined via KVL:

$$V_{Th} - I_{OUT}R_{Th} - V_{OUT} = 0$$
$$V_{Th} - 30\text{m} \times 50\text{m} - 3.298 = 0$$
$$\therefore V_{Th} = 3.300\text{ V}$$

Part h) [6 Marks]



Determine the maximum and minimum allowable values of $V_{in\text{-}}$, $V_{in\text{-},max}$ and $V_{in\text{-},min}$.

$V_{in\text{-},max} =$	2.25 V
$V_{in\text{-},min} =$	1.5 V

This is a difference amplifier:

$$V_{out} = \frac{R_f}{R_i}(V_{in+} - V_{in\text{-}}) = \frac{47}{4.7}(2 - V_{in\text{-}})$$
$$= 20 - 10V_{in\text{-}}$$

When $V_{in\text{-}}$ increases, V_{out} decreases, and vice versa. So, $V_{in\text{-},max}$ will be associated with the *minimum* allowable output voltage (−2.5 V) and $V_{in\text{-},min}$ will be associated with the maximum allowable output voltage (+5 V):

$$V_{out,min} = -2.5 = 20 - 10V_{in\text{-},max}, \therefore V_{in\text{-},max} = 2.25\text{ V}$$

$$V_{out,max} = 5 = 20 - 10V_{in\text{-},min}, \therefore V_{in\text{-},min} = 1.5\text{ V}$$

Part i) [5 Marks]

Lab Safety: Circle the most correct answer for each question.

1) What type of footwear is not allowed in the lab?

A. Sneakers
B. Running shoes
C. Sandals
D. Boots

2) Why is it dangerous to bring liquids into the lab?

A. They can cause fire or electrocution.
B. They could stain the lab manuals.
C. They can damage the breadboards.
D. It's just a general rule.

3) When preparing to measure current using a multimeter, which configuration is safe and correct?

A. Connect the meter in parallel with the powered circuit and set it to voltage mode.
B. Connect the meter directly across any charged capacitors and set it to resistance mode.
C. Connect the meter in parallel with the circuit component and set it to current mode.
D. Connect the meter in series with the powered circuit and set it to current mode.

4) You notice a frayed power cable on your lab power supply. What should you do?

A. Wrap the frayed cable securely with electrical tape and continue using it.
B. Inform your lab instructor immediately and avoid using the equipment.
C. Switch the unit off periodically to prevent overheating.
D. Handle the cable carefully, avoiding direct contact with the frayed area.

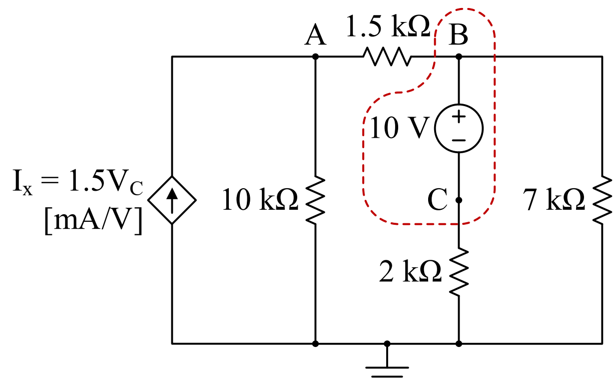
5) Before changing components in your circuit, you must first:

A. Lower the voltage slightly to avoid large current spikes.
B. Turn off the power supply and discharge all capacitors.
C. Ground the negative terminal of your power supply to the lab bench.
D. Confirm the circuit is functioning correctly to avoid misdiagnosing faults.

Question 2

[10 Marks]

Find the unknown node voltages and circle the supernode.



$V_A =$	20 V
$V_B =$	14 V
$V_C =$	4 V

$$1.5V_C = \frac{V_A}{10} + \frac{V_A - V_B}{1.5}$$

$$\frac{V_B - V_A}{1.5} + \frac{V_C}{2} + \frac{V_B}{7} = 0$$

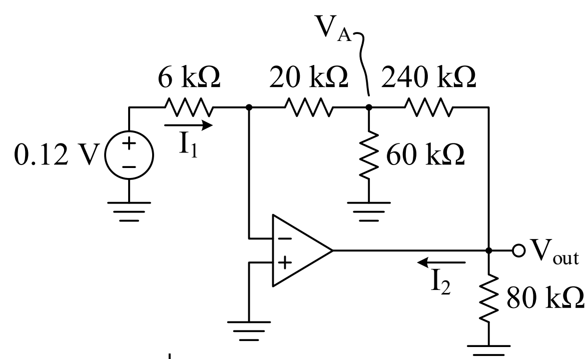
$$V_B = V_C + 10$$

Supernode in red

Question 3

[8 Marks]

Determine V_A, V_out, I_1, and I_2.



$V_\text{A} =$	−0.4 V
$V_\text{out} =$	−6.8 V
$I_1 =$	20 μA
$I_2 =$	112 μA

It's almost an inverting amplifier, but the feedback network looks too complicated to use $A = -R_\text{f}/R_\text{i}$. Since there's CLNF, start there. The opamp sets $V_- = V_+ = 0$ V. We can now do nodal at V_- and V_A:

KCL @ V_-:

$$\frac{V_- - 0.12}{6} + \frac{V_- - V_\text{A}}{20} = 0$$

Since $V_- = 0$ V:

$$-\frac{0.12}{6} - \frac{V_\text{A}}{20} = 0$$

$$\therefore V_\text{A} = -0.4 \text{ V}$$

KCL @ V_A:

$$\frac{V_\text{A} - V_-}{20} + \frac{V_\text{A} - 0}{60} + \frac{V_\text{A} - V_\text{out}}{240} = 0$$

$$\frac{0.4}{20} + \frac{0.4}{60} + \frac{0.4 + V_\text{out}}{240} = 0$$

$$\therefore V_\text{out} = -6.8 \text{ V}$$

$$I_1 = \frac{0.12 - 0}{6} = 20\ \mu\text{A}$$

KCL @ V_out to get I_2:

$$\frac{V_\text{out} - V_\text{A}}{240} + \frac{V_\text{out}}{80} + I_2 = 0$$

$$\frac{-6.8 - (-0.4)}{240} + \frac{-6.8}{80} + I_2 = 0$$

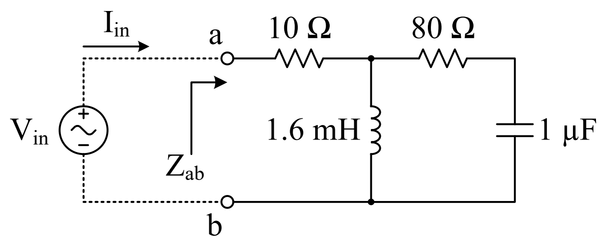
$$\therefore I_2 = 112\ \mu\text{A}$$

Question 4

[10 Marks]

Part a) [5 Marks]

Given $\omega_0 = 25\ \mathrm{krad/s}$, determine the impedances of L and C (Z_L, and Z_C, respectively), and the equivalent impedance at node a with respect to node b, Z_{ab}.



$Z_L =$	$j40\ \Omega = 40\angle 90°\ \Omega$
$Z_C =$	$-j40\ \Omega = 40\angle -90°\ \Omega$
$Z_{ab} =$	$30 + j40\ \Omega = 50\angle 53°\ \Omega$

$$Z_L = j\omega_0 L = j \times 25k \times 1.6m = j40\ \Omega = 40\angle 90°\ \Omega$$

$$Z_C = \frac{1}{j\omega_0 C} = -\frac{j}{25k \times 1\mu} = -j40\ \Omega = 40\angle -90°\ \Omega$$

$$\begin{aligned}
Z_{ab} &= 10 + [Z_L \parallel (80 + Z_C)] \\
&= 10 + [j40 \parallel (80 - j40)] \\
&= 10 + \left(\frac{1}{j40} + \frac{1}{80 - j40}\right)^{-1} \\
&= 10 + \left(\frac{80 - j40 + j40}{j40(80 - j40)}\right)^{-1} \\
&= 10 + \frac{j40(80 - j40)}{80} \\
&= 10 + j20(2 - j) \\
&= 10 + j40 + 20 \\
&= 30 + j40\ \Omega = 50\angle 53°\ \Omega
\end{aligned}$$

Part b) [2 Marks]

The voltage applied to node a is $\boldsymbol{V}_{\mathbf{in}} = 10\angle 0°\ \mathrm{V_{RMS}}$. Determine the current entering the circuit, $\boldsymbol{I}_{\mathbf{in}}$.

$\boldsymbol{I}_{\mathbf{in}} =$	$0.12 - j0.16\ \mathrm{A} = 0.2\angle -53°\ \mathrm{A}$

$$\boldsymbol{I}_{\mathbf{in}} = \frac{\boldsymbol{V}_{\mathbf{in}}}{Z_{ab}} = \frac{10\angle 0°}{50\angle 53°} = 0.12 - j0.16\ \mathrm{A} = 0.2\angle -53°\ \mathrm{A}$$

Part c) [3 Marks]

Determine the real and reactive power at the input, P_{in} and Q_{in}, respectively. Then, express this as a complex power phasor, $\boldsymbol{S}_{\mathbf{in}}$. Be sure to include the correct units.

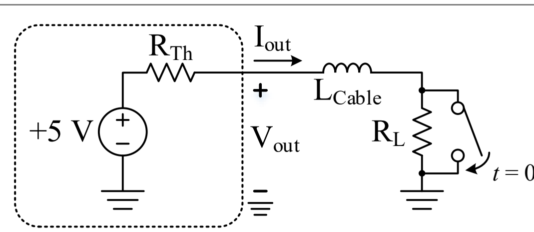
$P_{in} =$	$1.2\ \mathrm{W}$
$Q_{in} =$	$1.6\ \mathrm{VAR}$
$\boldsymbol{S}_{\mathbf{in}} =$	$1.2 + j1.6\ \mathrm{VA} = 2\angle 53°\ \mathrm{VA}$

$$\boldsymbol{S}_{\mathbf{i}} = \boldsymbol{V}_{\mathbf{in}}\boldsymbol{I}_{\mathbf{in}}^{*} = 10(0.12 + j0.16) = 1.2 + j1.6\ \mathrm{VA} = 2\angle 53°\ \mathrm{VA}$$

Question 5

[10 Marks]

The Thevenin equivalent circuit of a 5 V USB charger output is shown in the dashed box of Figure a). A long cable with inductance L_{Cable} connects the charger output to a load, R_L.



Part a) [4 Marks]

$R_{Th} =$	$50\ \Omega$
$I_{out} =$	$497.5\ \mu A$

Ignore the switch for now. When no load is connected, $V_{out} = 5.000\ V$. However, V_{out} stabilizes at 4.975 V when a load $R_L = 10\ k\Omega$ is connected. Determine R_{Th} and I_{out}.

$$V_{out} = V_{Th}\frac{R_L}{R_L + R_{Th}}, \therefore R_{Th} = \frac{V_{Th}R_L}{V_{out}} - R_L = \frac{5 \times 10k}{4.975} - 10k = 50\ \Omega$$

$$I_{out} = \frac{V_{out}}{R_L} = \frac{4.975}{10k} = 497.5\ \mu A$$

Part b) [6 Marks]

L_{Cable} is measured to be 4 µH. A short occurs at time $t = 0$, modelled as a switch closing around R_L. Determine an expression for the output current, $I_{out}(t)$, valid for $t \geq 0$. Determine $V_{out}(t = 100\ ns)$

$I_{out}(t) =$	$100 - 99.5e^{-\frac{t}{80n}}$ [mA]
$V_{out}(t = 100\ ns) =$	1.425 V

$$I_i = 497.5\ \mu A$$

$$I_f = I_{sc} = \frac{V_{Th}}{R_{Th}} = \frac{5}{50} = 100\ mA$$

$$\tau = \frac{L_{Cable}}{R_{Th}} = \frac{4\mu}{50} = 80\ ns$$

$$\therefore I_{out}(t) = I_f + (I_i - I_f)e^{-\frac{t}{\tau}} = 100 - 99.5e^{-\frac{t}{80n}}\ [mA]$$

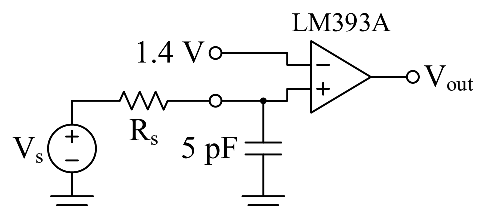
Given this current, the output voltage is determined by the drop across R_{Th}:

$$I_{out}(t = 100\ ns) = 100 - 99.5e^{-\frac{100n}{80n}}\ [mA] = 71.5\ mA$$

$$V_{out}(t = 100\ ns) = V_{Th} - I_{out}(t = 100\ ns)R_{Th} = 5 - 71.5m \times 50 = 1.425\ V$$

Question 6

[6 Marks]



A sensor generates a step output voltage signal, $V_s = 0 \rightarrow 5\ V$ when a dangerous material falls during handling. The system must be immediately disabled when V_s exceeds 1.4 V **with no more than a 3 µs delay**. An opamp comparator (LM393A) ensures there is minimal output delay once $V_+ > V_-$, however, the sensor's output resistance, R_s, interacts with the comparator's 5 pF input capacitance to cause an RC switching delay.

$R_{s,max} =$	$1.83\ M\Omega$

Assuming that the sensor triggers ($V_s = 0 \rightarrow 5\ V$) at $t = 0$, determine the maximum allowable sensor output resistance, $R_{s,max}$, so that the noninverting input voltage $V_+ =$ 1.4 V at t = 3 µs. Anything larger than this resistance causes a time constant that is too slow.

$$V_i = 0\ V, V_f = 5\ V, \tau = R_S C_+$$

$$V_+(t) = V_f + (V_i - V_f)e^{-\frac{t}{R_s C_+}} = 5 - 5e^{-\frac{t}{R_s 5p}}$$

$$\therefore 1.4 = 5 - 5e^{-\frac{3\mu}{R_{s,max} \times 5p}}$$

$$e^{-\frac{600k}{R_{s,max}}} = 0.72, \therefore R_{s,max} = -\frac{600k}{ln\ 0.72} = 1.83\ M\Omega$$